

CLIMATE LITERACY ACADEMY

# Academy catalogue

Translation draft from Turkish; not yet reviewed by a qualified reader. Design stage; not piloted.
The date, length, language and format of the Antalya session are still awaiting confirmation from the organisers.

The Climate Literacy Academy is an education model that starts with a pilot for ages 7–12 and extends to lifelong learning. Participants make sensory observations, explore a digital or printed scenario, collect evidence through a safe activity and follow up on a practical action. Teacher packs include sources, assessment guidance and accessible alternatives. Pilot outcomes have not yet been measured; partnerships and funding remain subject to confirmation.

## Learning cycle

- Observe: look, listen, touch
- Measure and record: keep observations separate from measurements
- Explain: state your evidence and its limits
- Try and track: follow one small action for 7, 30 and 90 days

## Who is it for?

**Teachers**
The Academy offers teachers and facilitators a sourced, traceable set of lessons. Each demonstration module includes a timed lesson plan, materials and quantities, a worksheet, cut-out cards, an answer key, a rubric, safety rules and equivalent ways to take part. The designs have not yet been tested in practice, and the experiments should be rehearsed before any pilot.

**Children**
Explore the climate with İKO! Look with your eyes, listen with your ears, touch with your hands, then measure and think. Tell the difference between what happens today and what has happened for many years. Choose one small action and follow it for a week. Smelling and tasting are optional; if you prefer, you can do the same task with cards.

**Adult participants**
The adult programmes focus on decisions at home, at work and as consumers. In each session you choose an action that suits your own circumstances, write down your assumptions and follow the action for a short period. You do not need to share bills, income or health information. The adult modules are at the design stage and have not yet been delivered.

**International visitors**
The Climate Literacy Academy is an education model designed in Türkiye. It starts with a pilot for ages 7–12 and is meant to extend to lifelong learning. Participants observe, test their observations with measurements, explain their reasoning and follow up on a small action. The activities shown here have not yet been delivered in a pilot, and their outcomes have not been measured. Partnerships and funding are still to be confirmed.

The sources support the mechanisms and the teaching approach; they do not prove the effect of this programme. Targets are not results. Example data are labelled everywhere.

## Age groups

| Title | Length | Summary | Status |
|---|---|---|---|
| **Ages 5–6** | 6 sessions × 45 minutes, with an accompanying adult | Exploration through pictures, stories and large pieces. No reading, writing or arithmetic is required: children choose with pictures and talk it through with an adult. | Extension programme: design ready; requires separate trialling and review by an early-years teacher. |
| **Ages 7–9** | 12 sessions × 90 minutes (including a 10-minute break) | Entry level of the pilot. Comparing two options, drawing, counting and giving a one-sentence reason, with an educator helping with measurements. | Pilot level: design ready; not yet tested in practice. |
| **Ages 10–12** | 12 sessions × 90 minutes (including a 10-minute break) | Measurement and reasoning level of the pilot. Predict–observe–explain, comparisons that change one variable, and expressing uncertainty. | Pilot level: design ready; not yet tested in practice. |
| **Ages 13–15** | 6 sessions × 100 minutes (including a 10-minute break) | Building cause-and-effect maps, weighing competing explanations, controlling variables and questioning sources. | Extension programme: design ready; not yet trialled or delivered. |
| **Ages 16–18** | 6 sessions × 110 minutes (including a 10-minute break) | Setting design criteria, repeat measurements, cost, uncertainty and ethics, leading to a small project. | Extension programme: design ready; not yet trialled or delivered. |
| **Ages 19–25** | 6 sessions × 120 minutes (including a 10-minute break) | For students, workers, job seekers and volunteers; no diploma required. Proposals with a functional unit, budget, task sharing and stated limits. | Extension programme: design ready; not yet trialled or delivered. |
| **Ages 26–59** | 6 sessions × 90 minutes (including a 10-minute break); evenings or weekends, with a printed option | Household, workplace and consumer decisions. Money, time, comfort and maintenance effort are weighed together; sharing personal bills is not required. | Extension programme: design ready; not yet trialled or delivered. |
| **Ages 60 and over** | 6 sessions × 75 minutes (including a 10-minute break) | An accessible pace, sharing of experience and intergenerational work. Personal experience is recorded separately from measurements. Age alone is not an indicator of ability. | Extension programme: design ready; not yet trialled or delivered. |

## Module summaries (60)

Only the summary of this module has been translated. The detailed teacher pack is currently available in Turkish only. Designed but not yet tested in practice. Learning outcomes have not been measured.

### Ages 5–6

| ID | Title | Length | Summary | Status |
|---|---|---|---|---|
| 05_06-M01 | From weather to climate | 45 min | Choosing what a puppet should wear today and saying why that choice does not hold for the whole year. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 05_06-M02 | Designing shade and cool spaces | 45 min | Making shade for a toy waiting area and drawing the shadow without looking at the sun. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 05_06-M03 | The journey of water and how we use it | 45 min | Spotting the tap left running unnecessarily in a hand-washing story while keeping every hygiene step. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 05_06-M04 | Food and portions with the five senses | 45 min | Choosing one's own small portion and using the right to say no; an adult weighs the leftovers. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 05_06-M05 | A second life for materials | 45 min | Making a reusable task-card box from cardboard. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 05_06-M06 | Trying out and tracking an action | 45 min | Choosing, with an adult, a three-day action of putting things back or caring for something. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |

### Ages 7–9

| ID | Title | Length | Summary | Status |
|---|---|---|---|---|
| 07_09-M01 | From weather to climate | 90 min | Sorting four cards into "today" and "many years" and explaining why one week of observation does not show climate change. | Detailed content: translation draft *demo_modules.07_09-M01* |
| 07_09-M02 | Designing shade and cool spaces | 90 min | Comparing two options for a resting spot in the academy garden. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |

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|---|---|---|---|---|
| 07_09-M03 | The journey of water and how we use it | 90 min | Using water tokens to make choices that protect hygiene. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 07_09-M04 | Soil cover and water | 90 min | Changing only the soil cover between two trays and measuring and drawing the water collected. | Detailed content: translation draft *demo_modules.07_09-M04* |
| 07_09-M05 | Food and portions with the five senses | 90 min | Comparing regular and odd-looking slices of the same apple variety with the senses and choosing a small portion. | Detailed content: translation draft *demo_modules.07_09-M05* |
| 07_09-M06 | A second life for materials | 90 min | Creating a collage about a climate observation and recording the second life of each piece. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 07_09-M07 | Making energy visible | 90 min | Reducing the time tokens of a model light left on unnecessarily. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 07_09-M08 | Living things and the web of care | 90 min | Matching three care needs in a tree nursery with picture cards. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 07_09-M09 | Colour from leaves and a materials account | 90 min | Comparing a dry-leaf rubbing with a colour trace made from spinach. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 07_09-M10 | Everyday routes and transport choices | 90 min | Planning a safe, shaded route to an imaginary school. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 07_09-M11 | Investigating a climate claim | 90 min | Explaining that a “cold today” card and a long-term graph answer different questions. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 07_09-M12 | Trying out and tracking an action | 90 min | Planning a one-week class action in three pictures. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |

## Ages 10–12

| ID | Title | Length | Summary | Status |
|---|---|---|---|---|

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|---|---|---|---|---|
| 10_12-M01 | From weather to climate | 90 min | Classifying eight statements by time scale and explaining the difference between daily observation and a climate normal. | Detailed content: translation draft *demo_modules.10_12-M01* |
| 10_12-M02 | Designing shade and cool spaces | 90 min | Taking three repeat measurements on the same surface and recording measurements separately from feelings. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 10_12-M03 | The journey of water and how we use it | 90 min | Calculating litres per minute for an example flow of 1 litre in 10 seconds and marking it as an example value. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 10_12-M04 | Soil cover and water | 90 min | Recording runoff volume and turbidity in a controlled tray comparison and explaining the limits of the experiment. | Detailed content: translation draft *demo_modules.10_12-M04* |
| 10_12-M05 | Food and portions with the five senses | 90 min | Recording edible apple leftovers under two serving conditions and discussing other possible causes of any difference. | Detailed content: translation draft *demo_modules.10_12-M05* |
| 10_12-M06 | A second life for materials | 90 min | Recording the mass of materials in a collage and how added glue affects later sorting. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 10_12-M07 | Making energy visible | 90 min | Explaining the 0.05 kWh calculation for an example LED. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 10_12-M08 | Living things and the web of care | 90 min | Equalising observation time in two areas and setting limits on conclusions about species richness. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 10_12-M09 | Colour from leaves and a materials account | 90 min | Recording the area coloured with 20 mL of water and how it changes the next day. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 10_12-M10 | Everyday routes and transport choices | 90 min | Building a table of time, accessibility and shade for three routes and discussing why there is no single right route. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 10_12-M11 | Investigating a climate claim | 90 min | Finding the institution, date and geographical coverage of a source. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |

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|---|---|---|---|---|
| 10_12-M12 | Trying out and tracking an action | 90 min | Writing an action plan with a baseline measurement and a day-30 check question. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |

## Ages 13–15

| ID | Title | Length | Summary | Status |
|---|---|---|---|---|
| 13_15-M01 | From weather to climate | 100 min | Testing the time scale in a social-media claim against long-term data. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 13_15-M02 | Soil cover and water | 100 min | Discussing alternative explanations and reproducibility in a controlled tray experiment. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 13_15-M03 | Food and portions with the five senses | 100 min | Comparing a small-portion-with-second-helping scenario for a school canteen. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 13_15-M04 | A second life for materials | 100 min | Designing a repairable pencil holder and discussing how new glue can reduce the benefit. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 13_15-M05 | Investigating a climate claim | 100 min | Breaking the claim “natural products are always eco-friendly” down into criteria. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 13_15-M06 | Trying out and tracking an action | 100 min | Planning a 30-day school-club task that creates no access barriers. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |

## Ages 16–18

| ID | Title | Length | Summary | Status |
|---|---|---|---|---|
| 16_18-M01 | Designing shade and cool spaces | 110 min | Designing a shade prototype and maintenance plan for a youth-centre waiting area. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 16_18-M02 | Soil cover and water | 110 min | Explaining why the soil-cover model cannot be transferred directly to the scale of a tree nursery. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |

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|---|---|---|---|---|
| 16_18-M03 | Food and portions with the five senses | 110 min | Proposing portion choices without changing the menu and defining consent for data collection. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 16_18-M04 | Making energy visible | 110 min | Calculating reductions in appliance use time while maintaining function and comfort. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 16_18-M05 | Investigating a climate claim | 110 min | Identifying the system boundary and missing evidence in a product's green claim. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 16_18-M06 | Trying out and tracking an action | 110 min | Developing a small prototype, assessed against three criteria, into a second version. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |

## Ages 19–25

| ID | Title | Length | Summary | Status |
|---|---|---|---|---|
| 19_25-M01 | Designing shade and cool spaces | 120 min | Preparing user requirements and a shade proposal for a campus or workplace waiting area. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 19_25-M02 | The journey of water and how we use it | 120 min | Observing water flow and duration in a shared kitchen while protecting hygiene and without requesting private consumption data. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 19_25-M03 | Food and portions with the five senses | 120 min | Designing a grams-per-person-per-meal measurement and a menu-based comparison plan for a canteen. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 19_25-M04 | Making energy visible | 120 min | Writing down the assumptions behind a low-cost proposal to change usage hours in a workspace. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 19_25-M05 | Investigating a climate claim | 120 min | Distinguishing verified data, estimates and marketing claims in a climate start-up. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |

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|---|---|---|---|---|
| 19_25-M06 | Trying out and tracking an action | 120 min | Proposing a small project with a lead person, budget, data and a stopping condition. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |

## Ages 26–59

| ID | Title | Length | Summary | Status |
|---|---|---|---|---|
| 26_59-M01 | From weather to climate | 90 min | Explaining, with one's own example, the difference between preparing for the day's weather and making long-term decisions at home or work. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 26_59-M02 | Designing shade and cool spaces | 90 min | Discussing shade, ventilation and maintenance options at home or work under safe conditions, without giving medical advice. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 26_59-M03 | The journey of water and how we use it | 90 min | Choosing one water use in a home or work routine whose duration could be reduced; sharing personal bills is not required. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 26_59-M04 | Food and portions with the five senses | 90 min | Planning a shopping list and small portions according to household size. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 26_59-M05 | Making energy visible | 90 min | Calculating kWh from an example appliance's use time; if a price is used, stating its date and tariff scope. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 26_59-M06 | Trying out and tracking an action | 90 min | Choosing a one-week trial that takes time, cost and maintenance effort into account. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |

## Ages 60 and over

| ID | Title | Length | Summary | Status |
|---|---|---|---|---|
| 60_plus-M01 | From weather to climate | 75 min | Comparing a past weather memory with dates and an official long-term record, without treating the memory as data. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 60_plus-M02 | Designing shade and cool spaces | 75 min | Creating a shade and access map for a seating area; going outside in hot weather is not required. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |

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|---|---|---|---|---|
| 60_plus-M03 | The journey of water and how we use it | 75 min | Developing a suggestion to cut unnecessary time at an easily reached point of water use. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 60_plus-M04 | Food and portions with the five senses | 75 min | Discussing traditional portion and storage practices against food-safety principles; an old method is not automatically safe. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 60_plus-M05 | A second life for materials | 75 min | Designing a durable everyday object from fabric or cardboard together with a younger participant. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |
| 60_plus-M06 | Trying out and tracking an action | 75 min | Choosing a small, measurable action while sharing a care experience with young people. | Summary only: translation draft — *Only the title and a one-sentence summary are translated. Detailed content exists in Turkish only (see detail_tr).* |

# Demonstration modules: detailed content

## 07_09-M01 · From weather to climate · Ages 7–9 · 90 minutes

> Designed but not yet tested in practice. Learning outcomes have not been measured. Translation draft: awaiting review by a language specialist.

**Goal (teacher):** Children distinguish “What is today like?” from “What is it usually like over many years?”, place at least 3 of 4 cards in the right basket and say that one week does not show climate change.

### Timed plan

- 0–10: Opening question and prediction with İKO
- 10–25: Sensory exploration at the window or in the garden; the educator reads the thermometer
- 25–40: Time Detective card game
- 40–50: Break
- 50–70: Today’s log entry and the EXAMPLE DATA table; who made the graph and for which years
- 70–82: Explaining the greenhouse effect and choosing a small action
- 82–90: Exit question

**Goal (participant):** Tell today’s weather apart from the climate of many years.

### Steps

- Look at the sky, listen to the sound and feel the air from the fan.
- Write the number on the thermometer and your own feeling separately.
- Put the cards in the TODAY and MANY YEARS baskets and say the word that tells you the time.
- Keep a weather diary at the same time each day for one week.

**Science note:** Standard climate normals are calculated over 30-year periods (WMO). According to the IPCC, 2011–2020 was about 1.1 °C warmer than 1850–1900.

### Five senses

| | |
|---|---|
| **Sight** | Draw or rate the clouds; read the thermometer display; find the institution, period and unit on the graph. |
| **Hearing** | Listen to a short rain and wind recording; listen to a classmate’s reasoning and summarise it. |
| **Touch** | Use a hand fan to send air onto your palm; write your feeling separately from the measurement. |
| **Smell** | Not used in this module. |
| **Taste** | Not used in this module. |

### Safety

- Do not look directly at the sun.
- Do not place the thermometer on a hot surface.
- No names, addresses or photos are collected.

**Assessment:** At least 3 of 4 cards correct with at least one reason; a long-term reason in the exit answer.

### Cards and answers

The JSON contains the main cards only; the full printed set is in the card PDF (Turkish only).

| | | |
|---|---|---|
| 07_09-M01-C01<br>**CARD 1** | “It is raining today.” Which basket does this card go in? | Answer: TODAY (weather). Time word: “today”. |

| | | |
|---|---|---|
| 07_09-M01-C02<br>**CARD 2** | “Strong winds are expected tomorrow.” Which basket does this card go in? | Answer: TODAY / NEXT FEW DAYS (weather forecast). “Tomorrow” is one day later; climate needs many years. |
| 07_09-M01-C03<br>**CARD 3** | “In this region, summers are usually dry over many years.” Which basket does this card go in? | Answer: MANY YEARS (climate pattern). Time words: “many years”, “usually”. |
| 07_09-M01-C04<br>**CARD 4** | “The July average for this city has been calculated from thirty years of records.” Which basket does this card go in? | Answer: MANY YEARS (climate normal). An official graph is shown only if the national meteorological data are verified (source to be verified); otherwise the card is discussed orally. |

**Visitor explanation:** Children aged 7–9 use a card game to tell weather from climate and learn that one week of observations does not show climate change. Design stage.

## 10_12-M01 · From weather to climate · Ages 10–12 · 90 minutes

> Designed but not yet tested in practice. Learning outcomes have not been measured. Translation draft: awaiting review by a language specialist.

**Goal (teacher):** Students classify eight statements as observation, forecast, climate or faulty inference, write the time scale and name one limitation of a measurement.

### Timed plan

- 0–10: The claim “Yesterday was cold, so there is no warming”; prediction
- 10–25: Measurements at two shaded spots; feelings in a separate column
- 25–40: Time Detective+ (8 cards, 4 categories)
- 40–50: Break
- 50–70: Average of the EXAMPLE DATA compared with the official graph
- 70–82: Greenhouse effect; a two-sentence reply to the claim; action
- 82–90: Exit question

**Goal (participant):** Find the time scale a statement refers to and spot faulty inferences.

### Steps

- Leave the thermometer in the shade for 3 minutes, then read it; note the sensor type, place and time.
- Sort the eight cards into four categories and write the time scale.
- Explain why the average of the EXAMPLE DATA is not a climate normal.
- Reply to the claim “Yesterday was cold, so there is no warming” in two sentences.

**Science note:** Standard climate normals are calculated over 30-year periods (WMO). According to the IPCC, 2011–2020 was about 1.1 °C warmer than 1850–1900.

### Five senses

| | |
|---|---|
| **Sight** | Draw or rate the clouds; read the thermometer display; find the institution, period and unit on the graph. |
| **Hearing** | Listen to a short rain and wind recording; listen to a classmate’s reasoning and summarise it. |
| **Touch** | Use a hand fan to send air onto your palm; write your feeling separately from the measurement. |
| **Smell** | Not used in this module. |
| **Taste** | Not used in this module. |

### Safety

- Do not look directly at the sun.
- Do not place the thermometer on a hot surface.
- No names, addresses or photos are collected.

**Assessment:** At least 6 of 8 cards correct with time scale; Card 7 justified as a faulty inference; at least one limitation of the measurement.

### Cards and answers

The JSON contains the main cards only; the full printed set is in the card PDF (Turkish only).

| | | |
|---|---|---|
| 10_12-M01-C01<br>**CARD 1** | “It is raining today.” Category and time scale? | Weather observation; hours/day. |

| | | |
|---|---|---|
| 10_12-M01-C02<br>**CARD 2** | “Strong winds are expected tomorrow.” Category and time scale? | Weather forecast; day. |
| 10_12-M01-C03<br>**CARD 3** | “In this region, summers are usually dry over many years.” | Climate (pattern); many years. |
| 10_12-M01-C04<br>**CARD 4** | “The July average for this city has been calculated from thirty years of records.” | Climate (normal); 30 years. Official graph: source to be verified; no graph is invented. |
| 10_12-M01-C05<br>**CARD 5** | “Last week we measured over 30 °C in the school yard every day at noon.” | Weather observation; one week, one location, one time of day. |
| 10_12-M01-C06<br>**CARD 6** | “According to long-term records, winters in this city are wetter than summers.” | Climate (seasonal pattern). EXAMPLE STATEMENT: do not use until checked against the city’s official data (source to be verified). |
| 10_12-M01-C07<br>**CARD 7** | “Yesterday was very cold, so there is no global warming.” | Faulty inference: a global, long-term conclusion drawn from one local day. |
| 10_12-M01-C08<br>**CARD 8** | “According to the IPCC, global surface temperature in 2011–2020 was about 1.1 °C higher than in 1850–1900.” | Climate (long-term change); a comparison spanning about 150 years [K01]. |

**Visitor explanation:** Students aged 10–12 sort statements by time scale and give reasoned replies to faulty inferences such as “yesterday was cold, so there is no warming”. Design stage.

## 07_09-M04 · Soil cover and water · Ages 7–9 · 90 minutes

> Designed but not yet tested in practice. Learning outcomes have not been measured. Translation draft: awaiting review by a language specialist.

**Goal (teacher):** Children say that only the cover changes, record the water (mL) and its appearance in both collecting jugs, and explain that the model cannot give a definitive result for a field.

**Timed plan**

- 0–10: Prediction
- 10–25: Examining straw and soil with gloves
- 25–40: Rain Designer game (not experimental data)
- 40–50: Break
- 50–70: Experiment with the standard protocol: two cups poured at once, 3-minute wait, reading
- 70–82: Class results: in how many teams did the covered tray's jug collect less water?
- 82–90: Exit question and clean-up

**Goal (participant):** Does covering the soil change the water that reaches the jug?

**Steps**

- Tick your prediction.
- Name what we keep the same: soil, height of the back edge, amount of water, cup, waiting time.
- Pour the two cups at the same time with a partner and wait 3 minutes.
- Read the water in the jugs, compare its colour with the card and draw it.

**Standard protocol:** All teams use the same protocol: same mass of soil, 50 mL pre-wetting, dry straw only (25 g), back edge of the tray raised by 3 cm (if a different height is chosen during the rehearsal, all teams use that same height), two identical rain cups (100 mL each, poured together), 3-minute wait.

**Science note:** According to FAO, organic soil cover can reduce evaporation and erosion and help water soak into the soil. The tray is only a small model.

**Five senses**

| | |
|---|---|
| **Sight** | Read the water line in the collecting jug at eye level; rate cloudiness from 1 to 3 with the "READ" card. |
| **Hearing** | Count the seconds or time the pour while the rain cup empties. |
| **Touch** | With gloves, or through a closed bag, compare dry straw with dry soil. |
| **Smell** | Not used: soil is never smelled. |
| **Taste** | Not used. |

**Safety**

- Wear gloves and wash your hands with soap at the end.
- Never smell or taste the soil, and keep it away from your mouth and eyes.
- Do not drink the experiment water; wipe up spills straight away.

**Assessment:** At least 3 fixed conditions; two measurements (mL) and cloudiness; stating a limit of the model.

**Cards and answers**

The JSON contains the main cards only; the full printed set is in the card PDF (Turkish only).

| | | |
|---|---|---|
| 07_09-M04-C02<br>**COVER: DRY STRAW** | There is dry straw on top of the soil. Will the jug collect a little or a lot of water? | This is the only cover used in the experiment. The prediction can be compared with the result afterwards. |

| | | |
|---|---|---|
| 07_09-M04-C08<br>**CLOUDINESS CARD** | Hold the card behind the jug and look at the text: 1 I can read it easily · 2 It is hard to read · 3 I cannot read it | Scale: 1 clear, 2 slightly cloudy, 3 very cloudy. Use the same card for both jugs. |

**Game note:** The cover and rain scenarios in the game (leaves, gravel, plastic, heavy rain) are for discussion only; they are not tested and are not recorded as experimental data.

**Visitor explanation:** Children aged 7–9 change only the soil cover between two trays, measure the water that runs off and discuss the limits of a small model. Design stage.

## 10_12-M04 · Soil cover and water · Ages 10–12 · 90 minutes

> Designed but not yet tested in practice. Learning outcomes have not been measured. Translation draft: awaiting review by a language specialist.

**Goal (teacher):** Students name the variables, record runoff volume and cloudiness, compare team and class results, and write a reason and two limitations. Calculating the median is an extension task.

### Timed plan

- 0–10: Research question and prediction
- 10–25: Listing variables and reading the protocol
- 25–40: Rain Designer (not experimental data)
- 40–50: Break
- 50–70: Experiment with the standard protocol and class table
- 70–82: Number of teams with results in the same direction; reasons and limitations; optional median
- 82–90: Exit question and clean-up

**Goal (participant):** Set up a controlled comparison: change only the cover and measure the water collected.

### Steps

- Write the independent variable, what you measure and at least four fixed conditions.
- Record runoff volume (mL), cloudiness and cup timings; note any deviation.
- Compare your team's two results; count how many teams got results in the same direction.
- Write a reason, an alternative explanation and two limitations.

**Standard protocol:** All teams use the same protocol: same mass of soil, 50 mL pre-wetting, dry straw only (25 g), back edge of the tray raised by 3 cm (if a different height is chosen during the rehearsal, all teams use that same height), two identical rain cups (100 mL each, poured together), 3-minute wait, readings rounded to 5 mL. "100 minus runoff" is not treated as a measurement.

**Science note:** According to FAO, organic soil cover can reduce evaporation and erosion and help water soak into the soil. The tray is a small model and gives no field-scale result.

### Five senses

| | |
|---|---|
| **Sight** | Read the water line in the collecting jug at eye level; rate cloudiness from 1 to 3 with the "READ" card. |
| **Hearing** | Count the seconds or time the pour while the rain cup empties. |
| **Touch** | With gloves, or through a closed bag, compare dry straw with dry soil. |
| **Smell** | Not used: soil is never smelled. |
| **Taste** | Not used. |

### Safety

- Wear gloves and wash your hands with soap at the end.
- Never smell or taste the soil, and keep it away from your mouth and eyes.
- Do not drink the experiment water; wipe up spills straight away.

**Assessment:** Variable table; two runoff measurements and cloudiness; within-team and class comparison; a reason and two limitations. Median and range are extension tasks.

### Cards and answers

The JSON contains the main cards only; the full printed set is in the card PDF (Turkish only).

| | | |
|---|---|---|
| 10_12-M04-C02<br>**COVER: DRY STRAW** | A fibrous, porous layer. Your prediction for runoff and cloudiness? | The only cover used in the experiment. Mechanism: less raindrop impact, absorption/holding. |
| 10_12-M04-C12<br>**CLOUDINESS CARD** | Hold the card behind the jug: 1 Clear (text readable) · 2 Slightly cloudy · 3 Very cloudy (unreadable) | Use the same card for both jugs; read at eye level. |

**Game note:** The cover and rain scenarios in the game (leaves, gravel, plastic, heavy rain) are for discussion only; they are not tested and are not recorded as experimental data.

**Visitor explanation:** Students aged 10–12 apply the same protocol across five teams, compare runoff and cloudiness, and write down the limits of the experiment. Design stage.

## 07_09-M05 · Food and portions with the five senses · Ages 7–9 · 90 minutes

> Designed but not yet tested in practice. Learning outcomes have not been measured. Translation draft: awaiting review by a language specialist.

**Goal (teacher):** Children say that appearance, liking and safety are different things, choose a small portion and watch their team's leftovers being weighed. Tasting is optional; the card option and the data-keeper role count just the same.

**Timed plan**

- 0–10: Question comparing two apples; choice of how to take part
- 10–25: Hand-washing; look, touch, smell, listen, taste (optional)
- 25–40: Portion Chef card game
- 40–50: Break
- 50–70: Small portion (1 Sample 1 + 1 Sample 2), more on request; weighing the team's apple leftovers; carrot recorded separately
- 70–82: Explanation and action
- 82–90: Exit question and clean-up

**Goal (participant):** Compare two apples that look different, and take a little first, then more if you want.

**Steps**

- Wash your hands and choose how to take part: tasting, smelling only, cards or data keeper.
- Look at and touch Sample 1 and Sample 2; if you wish, smell, listen and taste.
- Take a small portion; ask for 2 more slices if you want.
- Write down your team's leftovers in grams.

**Science note:** According to UNEP, about 1.05 billion tonnes of food were wasted worldwide in 2022 (including inedible parts), about 60% of it in households. This figure does not translate directly to Türkiye or to this lesson.

**Five senses**

| | |
|---|---|
| **Sight** | Compare the colour and shape of the regular-looking Sample 1 and odd-looking Sample 2 slices. |
| **Hearing** | If you wish, listen to your own crunch, or to a recording. |
| **Touch** | With clean hands or a fork, compare how firm the slices are. |
| **Smell** | Optional: smell the slice on your own plate from about 10 cm. Smell does not show that food is safe. |
| **Taste** | Optional: taste slices that adults have checked in advance, and note your preference. Taste does not show whether food is safe or fit to eat. |

**Safety**

- Parental consent and allergy checks come before any tasting.
- Everyone uses their own plate and fork and eats sitting down.
- Smelling and tasting are optional; the card option counts just the same.
- Nobody is pressured to finish their plate.

**Assessment:** Choosing one's own portion; explaining that smell and taste do not show safety; recording the team's apple leftovers. Tasting is not a condition for success.

**Cards and answers**

The JSON contains the main cards only; the full printed set is in the card PDF (Turkish only).

| | | |
|---|---|---|
| 07_09-M05-C01<br>**GUEST 1** | Very hungry. Likes apples. How much will you serve? | Example plan: small portion + option to ask for more. |
| 07_09-M05-C02<br>**GUEST 2** | A little hungry. Trying apples for the first time and not sure. | Example plan: small portion, or try one slice. |
| 07_09-M05-C03<br>**GUEST 3** | Has just eaten and is full. | Example plan: small portion or none; pressure creates leftovers. |
| 07_09-M05-C04<br>**GUEST 4** | A little hungry. Likes apples but not sour ones. | Example plan: small portion; taste first. |
| 07_09-M05-C05<br>**SMALL PORTION** | 2 slices | Take a little first. |
| 07_09-M05-C06<br>**EXTRA** | If you want, 2 more slices | Extra slices are counted. |

**Visitor explanation:** Children aged 7–9 compare two differently shaped apples with their senses and try the “take a little first” approach. Tasting is optional and requires prior consent and allergy checks. Design stage.

## 10_12-M05 · Food and portions with the five senses · Ages 10–12 · 90 minutes

> Designed but not yet tested in practice. Learning outcomes have not been measured. Translation draft: awaiting review by a language specialist.

**Goal (teacher):** Students separate observation, liking, safety and environmental impact; record apple leftovers under two serving conditions; and write other possible causes of any difference. Grams per person are calculated per condition: total apple leftovers ÷ total number of people served apple in the portion stage. If that number is 0, the result is recorded as "cannot be calculated".

**Timed plan**

- 0–10: Research question; choice of how to take part
- 10–25: Sensory exploration; four columns
- 25–40: Portion Chef (fictional canteen)
- 40–50: Break
- 50–70: Standard condition (4 slices) and Small + extra condition (2 slices + more on request); weighing apple leftovers; carrot separate
- 70–82: Grams per person by condition (on the board); other causes
- 82–90: Exit question and clean-up

**Goal (participant):** Does taking a little first and adding more reduce the edible apple left on plates? Observe it in a small trial.

**Steps**

- Choose how to take part: tasting, smelling only, cards or data keeper.
- Compare Sample 1 and Sample 2 and fill in the four columns.
- Record how many people in your team were served apple in the portion stage (including anyone who ate none of it), the extra slices and the apple leftovers (g); record carrot separately.
- Write two other reasons that could explain a difference between the conditions.

**Science note:** According to UNEP, about 1.05 billion tonnes of food were wasted worldwide in 2022 (including inedible parts), about 60% of it in households. This figure does not translate directly to this lesson. No carbon calculation is made.

**Five senses**

| | |
|---|---|
| **Sight** | Compare the colour and shape of the regular-looking Sample 1 and odd-looking Sample 2 slices. |
| **Hearing** | If you wish, listen to your own crunch, or to a recording. |
| **Touch** | With clean hands or a fork, compare how firm the slices are. |
| **Smell** | Optional: smell the slice on your own plate from about 10 cm. Smell does not show that food is safe. |
| **Taste** | Optional: taste slices that adults have checked in advance, and note your preference. Taste does not show whether food is safe or fit to eat. |

**Safety**

- Parental consent and allergy checks come before any tasting.
- Everyone uses their own plate and fork and eats sitting down.
- Smelling and tasting are optional; the card option counts just the same.
- Nobody is pressured to finish their plate.

**Assessment:** Team data (people, extra slices, grams; carrot separate); explaining that smell and taste do not show safety; at least two other causes. Calculating independently is an extension task; tasting is not required.

**Calculation rule:** Denominator: number of people served apple in the portion stage (including anyone who did not touch their portion). Numerator: apple left on those plates (g), untouched portions included. People not served apple (those who did not taste, those who had carrot) are not counted. Leftovers from the sensory stage are excluded. If the number of people is 0, no division is made: “cannot be calculated”.

### Cards and answers

The JSON contains the main cards only; the full printed set is in the card PDF (Turkish only).

| | | |
|---|---|---|
| 10_12-M05-C04 **CONDITION: STANDARD** | Everyone gets a ready-made plate: 4 slices (2 Sample 1 + 2 Sample 2). | Served = people served apple × 4. |
| 10_12-M05-C05 **CONDITION: SMALL + EXTRA** | Everyone first gets 2 slices (1 Sample 1 + 1 Sample 2). Anyone who wants more takes 2 extra. | Served = people × 2 + extra slices. Extra slices are counted. |
| 10_12-M05-C10 **OBSERVATION · LIKING** | Observation: what I see, touch and hear. Liking: varies from person to person. | First two headings of the four-column table. |
| 10_12-M05-C11 **SAFETY · ENVIRONMENTAL IMPACT** | Safety: checks and hygiene. Smell and taste are not safety tests. Environmental impact: the resources used in production. | Last two headings of the four-column table. |

**Visitor explanation:** Students aged 10–12 compare two ways of serving in a small trial, weigh leftovers at team level and discuss the limits of the result. Design stage.